

Quantum walks for 1D anyons in a cavity

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I. INTRODUCTION

Los anyones son ...

Añadiendo el término de cavidad:

$$\mathcal{H}_{cav} = g_{cav} \sum_{i,j=1}^M (-1)^{i-j} \hat{n}_i \hat{n}_j = g_{cav} \left(\sum_{i=1}^M (-1)^i \hat{n}_i \right)^2, \quad (2)$$

Obteniendo el Hamiltoniano extendido:

$$\mathcal{H}_{ext}^b = \mathcal{H}^b + \mathcal{H}_{cav} \quad (3)$$

The original Hamiltonian (1) breaks the inversion symmetry $\mathcal{I}$, as the particles in the chain acquire a phase due a site occupation. Result, only for a change of sign in both term we will observe a total reflection in the quantum walk

II. MAIN IDEA

Del Hamiltoniano de bosones condicionados:

$$\mathcal{H}^b = -J \sum_{i=1}^M \left(\hat{b}_i^\dagger \hat{b}_{i+1} e^{i\theta \hat{n}_i} + h.c. \right) + \frac{U}{2} \sum_{i=1}^M \hat{n}_i (\hat{n}_i - 1), \quad (1)$$

$$\langle \hat{n}_j(t) \rangle_{+U,+g} = \langle \hat{n}'_j(t) \rangle_{-U,-g} \quad (4)$$

$$\langle \hat{n}_j(t) \rangle_{+U,-g} = \langle \hat{n}'_j(t) \rangle_{-U,+g} \quad (5)$$

III. CONCLUSION

Appendix: Appendix

Del operador de paridad:

$$\mathcal{P} = e^{i\pi \sum_r \hat{n}_{2r+1}} \quad (A.1)$$

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